

AI Intelligent Freight Quote Generation System

Dashboards, Login / RBAC, End-to-End Flow & M3 Test Flow

One Next.js application with role-based access control. Five major workspaces are recommended: User, Admin, Customs Compliance, AI Agent Operations, and Analytics / Management.

1. Final Dashboard Structure

```
FREIGHT QUOTE PLATFORM  
↓  
LOGIN / RBAC  
↓  
██████████████████████████████████████████████████████████████████  
USER ADMIN CUSTOMS AGENT ANALYTICS  
DASH DASH WORKSPACE CENTER /MANAGEMENT
```

1) User Dashboard — shipment, routes, quotes, documents. 2) Admin Dashboard — users, pricing, routes, integrations, settings. 3) Customs Compliance Workspace — reviews, documents, regulations, sign-off. 4) AI Agent Operations Center — Route/Pricing/Weather/Customs/Risk health and executions. 5) Analytics & Management — quote, pricing, risk, weather, customs, route and ML analytics.

Risk Dashboard, Weather Dashboard, Customs Dashboard and ML Dashboard are modules/pages inside these workspaces, not separate portals.

2. Login & RBAC Flow

```

Login
↓
Authentication API
↓
Credentials / token validation
↓
Load role + permissions
↓
Route guard
↓
USER → /user/dashboard
ADMIN → /admin/dashboard
CUSTOMS_OFFICER → /customs/dashboard
AGENT_OPERATOR → /agents/dashboard
MANAGER → /analytics/dashboard

```

Recommended roles: USER, ADMIN, CUSTOMS_OFFICER, AGENT_OPERATOR, MANAGER. Permissions must be checked separately from role names.

3. User Flow

```

graph TD
    A[User Login] --> B[Dashboard]
    B --> C[Create Shipment]
    C --> D[Validate origin / destination / cargo / HS code / Incoterm]
    D --> E[Route Intelligence]
    E --> F[Pricing Rule + ML]
    F --> G[Weather Agent]
    G --> H[Customs Agent]
    H --> I[Risk Engine]
    I --> J[Quote Preview]
    J --> K[Customs sign-off if required]
    K --> L[Quote Issued]
    L --> M[User views price + ETA + risk + compliance]

```

4. Admin Flow

Admin Login
↓
Admin Dashboard
■ ■ Users / Roles
■ ■ Shipments / Quotes
■ ■ Pricing Rules
■ ■ Routes / Carriers / Ports
■ ■ Risk / Customs Configuration
■ ■ Integrations / Data Freshness
■ ■ Alerts
■ ■ Audit Logs
■ ■ Settings

5. Customs Officer Flow

Customs Officer Login
↓
Customs Compliance Workspace
↓
Pending Reviews
↓
Open Compliance Case
↓
Review shipment + HS code + commodity + Incoterm
↓
Review required documents
↓
Review uploaded documents
↓
Review regulation evidence / citations
↓
Review AI risk findings
↓
APPROVE / REQUEST DOCUMENTS / CONDITIONAL / REJECT
↓
Audit Record
↓
Quote State Updated

If mandatory customs approval is not completed, the final quote remains HOLD/BLOCKED according to policy.

6. AI Agent Operations Center

Agent Operator / Authorized Admin
↓
AI Agent Operations Center
■ ■ Agent Overview
■ ■ Route Agent
■ ■ Pricing Agent
■ ■ Weather Agent
■ ■ Customs Agent
■ ■ Risk Engine
■ ■ Execution History

Each agent detail page shows status, requests, success/failure, latency, last execution, provider errors and recent executions.

Do not create five separate agent portals. Use one Agent Operations Center with agent detail pages.

7. Analytics / Management Flow

Manager / Authorized Admin
↓
Analytics Dashboard
■ ■ Executive Overview
■ ■ Quote Analytics
■ ■ Pricing Analytics
■ ■ Risk Analytics
■ ■ Weather Risk
■ ■ Customs Compliance
■ ■ Route Analytics
■ ■ ML Model Performance

8. Complete End-to-End Project Flow

```

Customer / Sales User
↓
Login
↓
Create Shipment
↓
Validate Data
↓
Routing
↓
Parallel Processing
■ Pricing
■ Weather Agent
■ Customs Agent
↓
Risk Engine
↓
Composite Risk + Explanation
↓
Customs Decision
■ Auto Pass → Continue
■ Human Review → Customs Officer
■ ↓
■ Approve / Docs / Reject
■ Missing Data → Hold
↓
Quote State Machine
↓
Quote Issuance
↓
User receives quote
↓
Audit + Alerts + Monitoring

```

9. Weather Test Flow

```

Create valid shipment → Generate route → Trigger weather assessment
→ Verify route geometry → Verify sampled points → Verify provider response
→ Verify observations saved → Verify storm detection → Verify risk score
→ Verify delay probability → Verify WeatherAssessment → Verify alert threshold
→ Verify quote shows weather risk.

```

Expected: every active route receives a weather-risk result or explicit degraded/fallback status.

10. Customs Test Flow

```

Create shipment with origin/destination/HS code/commodity/Incoterm
→ Validate customs → Retrieve regulations → Hybrid search/reranking
→ Generate required documents → Verify evidence/citations
→ Calculate readiness → Force missing mandatory document
→ Verify NEEDS_DOCUMENTS / NEEDS_REVIEW
→ Verify quote HOLD/BLOCKED
→ Login as Customs Officer → Open Pending Review
→ Review evidence/documents → Approve
→ Verify quote state changes → Verify audit record.

```

11. Customs Negative Tests

```

Invalid HS code → validation failure → no approval
Missing mandatory document → NEEDS_DOCUMENTS → quote held
Regulation source unavailable → approved fallback/cached data or degraded state
Officer rejects → REJECTED → quote blocked
Unauthorized user approves → 403 → no state change → security/audit event

```

12. Login & RBAC Tests

```

Valid USER → /user/dashboard
Valid ADMIN → /admin/dashboard
Valid CUSTOMS_OFFICER → /customs/dashboard
Valid AGENT_OPERATOR → /agents/dashboard
Valid MANAGER → /analytics/dashboard
Invalid password → rejected
USER opens /admin → 403
USER calls customs approval API → 403
Expired token → refresh/re-login
Logout → protected routes inaccessible

```

13. Quote State Test

DRAFT
↓
VALIDATED
↓
ROUTE_SELECTED
↓
PRICED
↓
WEATHER_ASSESSED
↓
CUSTOMS_ASSESSED
↓
RISK_ASSESSED
↓
Customs approval required?
■ NO → READY
■ YES → PENDING_REVIEW
■ APPROVED → READY
■ REQUEST_DOCUMENTS → HOLD
■ REJECTED → BLOCKED
↓
Risk policy check
↓
READY_FOR_ISSUANCE
↓
ISSUED

14. Agent Failure Test

Shipment → Agent → Provider timeout
→ Retry → If retry fails, use valid fallback/cache if available
→ Mark degraded state → Create monitoring event
→ Never silently generate false confidence.

Apply the same pattern to Route, Pricing, Weather, Customs and Risk services.

15. Dashboard Access Matrix

Workspace	USER	ADMIN	CUSTOMS	AGENT OP	MANAGER
User	✓	✓*	-	-	✓*
Admin	-	✓	-	-	✓*
Customs	-	✓*	✓	-	✓*
Agents	-	✓*	-	✓	✓*
Analytics	-	✓	-	✓*	✓

* Access should be permission-based, not automatically granted to every user of that role.

16. Frontend Route Structure

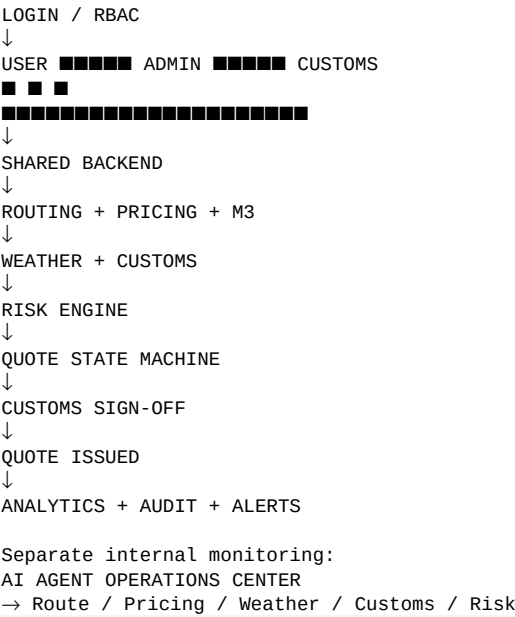
app/
■ login/
■ user/ → dashboard, shipments, routes, quotes, documents
■ admin/ → dashboard, users, roles, pricing, routes, integrations, audit
■ customs/ → dashboard, pending-reviews, compliance, documents, regulations
■ agents/ → dashboard, route, pricing, weather, customs, risk, executions
■ analytics/ → dashboard, quotes, pricing, risk, weather, customs, routes, ml

17. M3 Full Test Checklist

- All five login paths work
- Unauthorized routes and APIs return 403
- User can create shipment
- Route flow works
- Rule pricing works
- ML pricing comparison works
- Weather assessment works
- Weather risk appears in quote
- Customs validation works
- Regulation evidence/citations appear

- Customs checklist generated
- Missing-document flow works
- Customs Officer can review
- Officer approval changes quote state
- Officer rejection blocks quote
- Composite risk calculated
- Risk explanation visible
- Alerts generated
- Agent health visible
- Agent failures logged
- Data freshness visible
- Audit trail stored
- End-to-end quote flow passes

18. Final Architecture



Final recommendation: build ONE Next.js frontend with RBAC and five role-based workspaces. The agents are services inside the same quote-generation pipeline; the Agent Operations Center exists only to monitor and troubleshoot them.